

The 26th Hong Kong (China) Mathematical Olympiad

9 Dec 2023

Problem 1. Let $A_1A_2\cdots A_n$ be a convex polygon where $n \geq 4$. For each $i = 1, 2, \dots, n$, let $f(i)$ be the maximum of the angles $\angle A_iA_jA_{i+1}$ where A_j can be any vertex different from A_i and A_{i+1} (with $A_{n+1} = A_1$). For each n , find the smallest possible value of $f(1) + f(2) + \cdots + f(n)$.

Problem 2. Given an acute $\triangle ABC$ with $AB > AC$, let M be the midpoint of the side BC . Let I be the incentre of $\triangle ABC$, and let D be the foot of perpendicular from I to BC . The line passing through A and perpendicular to BC meets the circumcircle of $\triangle ABC$ again at $E \neq A$. Suppose $\angle MID = \angle ACB - \angle ABC$. Prove that $DE = DI$.

Problem 3. There are $2n$ cards labelled $1, 2, \dots, 2n$ respectively, where n is a positive integer. Two players A and B take turns choosing a card which has not been chosen by anyone, with A going first, until all cards are chosen. Let $a_1, a_2, \dots, a_n$ be the labels of the cards chosen by A in this order, and let $b_1, b_2, \dots, b_n$ be the labels of the cards chosen by B in this order. The score of A is equal to

$$(a_1 + a_2 + \cdots + a_n)^2 + (b_1 + b_2 + \cdots + b_n)^2 - (a_1b_1 + a_2b_2 + \cdots + a_nb_n).$$

What is the maximum score (in terms of n) that A can guarantee to obtain regardless of how B plays?

Problem 4. Find all positive integer(s) n such that $(2^n - 1)(5^n - 1)$ is a perfect square.

Time: 3 hours
Each problem is worth 7 points

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問題 1. 設 $A_1A_2\cdots A_n$ 是一個凸多邊形，其中 $n \geq 4$ 。對每個 $i = 1, 2, \dots, n$ ，設 $f(i)$ 是所有角 $\angle A_iA_jA_{i+1}$ 當中最大的一個，其中 A_j 可以是異於 A_i 和 A_{i+1} 的頂點（這裏 $A_{n+1} = A_1$ ）。對每個 n ，求 $f(1) + f(2) + \cdots + f(n)$ 的最小可能值。

問題 2. 給定銳角 $\triangle ABC$ ，其中 $AB > AC$ ，設 M 是邊 BC 的中點，設 I 是 $\triangle ABC$ 的內心，並設 D 是 I 到 BC 的垂足。穿過 A 並與 BC 垂直的直線與 $\triangle ABC$ 的外接圓再相交於 $E \neq A$ 。假設 $\angle MID = \angle ACB - \angle ABC$ 。求證： $DE = DI$ 。

問題 3. 現有 $2n$ 張分別編號為 $1, 2, \dots, 2n$ 的咭，其中 n 是正整數。兩個玩家甲和乙輪流取走一張未被任何人取走的咭（由甲開始），直至所有咭都被取走。設 $a_1, a_2, \dots, a_n$ 是甲依次取走的咭的編號，並設 $b_1, b_2, \dots, b_n$ 是乙依次取走的咭的編號。甲的得分為

$$(a_1 + a_2 + \cdots + a_n)^2 + (b_1 + b_2 + \cdots + b_n)^2 - (a_1b_1 + a_2b_2 + \cdots + a_nb_n)。$$

在無論乙如何操作的情況下，甲所能保證的得分的最大值是甚麼（答案以 n 表示）？

問題 4. 求所有正整數 n ，使得 $(2^n - 1)(5^n - 1)$ 是完全平方數。

考試時間：3 小時
每題 7 分

Solutions

1. For each $k = 2, 3, \dots, n-1$, define $S_k = \sum_{i=1}^n \angle A_i A_{i+k} A_{i+1}$ where the indices are taken modulo n . Note that

$$S_2 + S_3 + \dots + S_{n-1} = \angle A_1 + \angle A_2 + \dots + \angle A_n = (n-2) \cdot 180^\circ \quad (1)$$

because each angle $\angle A_i A_j A_k$ with distinct i, j, k is counted exactly once. By definition, the given sum S is at least S_k for each $k = 2, 3, \dots, n-1$. Therefore, by equation (1), we obtain

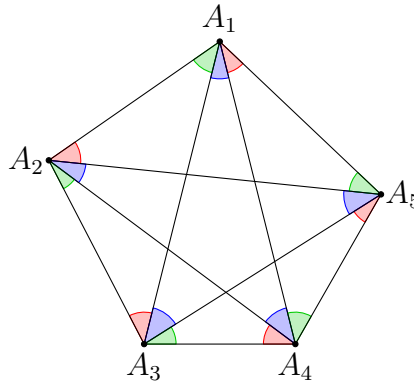
$$S \geq \frac{S_2 + S_3 + \dots + S_{n-1}}{n-2} = 180^\circ.$$

On the other hand, if $A_1 A_2 \dots A_n$ is cyclic (for example, it is a regular n -gon), then

$$f(1) + f(2) + \dots + f(n) = \widehat{A_1 A_2} + \widehat{A_2 A_3} + \dots + \widehat{A_n A_1} = 180^\circ.$$

Therefore, the smallest possible value of S is 180° .

(In the figure below, we have $n = 5$. Also, S_2 is the sum of all angles in red, S_3 is the sum of all angles in blue, and S_4 is the sum of all angles in green.)



2. Let O and H be the circumcentre and the orthocentre of $\triangle ABC$ respectively. Reflect A about O to a point A' . Reflect D about M and I to points D' and P respectively. Also, let Q be the midpoint of AH . Notice the following well-known results.

- E and H are symmetric about the line BC
- M is the midpoint of $A'H$
- A, P, D' are collinear (as D' is the contact point of the A -excircle and the side BC)
- $\angle OAI = \angle HAI$

$$\begin{aligned} N &:= \left(\sum_{k=1}^n 2k \right)^2 + \left(\sum_{k=1}^n (2k-1) \right)^2 - \sum_{k=1}^n (2k-1)(2k) \\ &= n^2(n+1)^2 + n^4 - \left(\frac{2n(n+1)(2n+1)}{3} - n(n+1) \right) \\ &= \frac{n(6n^3 + 2n^2 + 1)}{3}. \end{aligned}$$
$$\begin{aligned} X &= a_1 + a_2 + \cdots + a_n, \\ Y &= b_1 + b_2 + \cdots + b_n, \\ S &= X^2 + Y^2, \\ T &= a_1b_1 + a_2b_2 + \cdots + a_nb_n. \end{aligned}$$

Proof. Since there are finitely many possible ways to assign the values, T must attain a maximum value. For this maximal case, we may assume $a_i < b_i$ for each i . Note that we may assume $\{a_i\}$ and $\{b_i\}$ are increasing by the rearrangement inequality.

Under these assumptions, we must have $a_1 = 1$. We are going to show that $a_i = 2i - 1$ and $b_i = 2i$ for all i . Suppose on the contrary that this does not hold. Then there is a smallest index k such that $b_k \neq 2k$. Note that we must have $a_i = 2i - 1$ for $i = 1, 2, \dots, k$ and $a_{k+1} = 2k$ (clearly $k \neq n$). This implies $a_k < a_{k+1} < b_k < b_{k+1}$. By the rearrangement inequality, we have

$$a_k b_k + b_{k+1} a_{k+1} < a_k a_{k+1} + b_{k+1} b_k.$$

This means we can swap b_k and a_{k+1} to increase the value of T , contradiction.

Therefore, T is maximized when $a_i = 2i - 1$ and $b_i = 2i$ for all i . This proves the claim. $\square$

Now, we show that A can guarantee to obtain score N . A strategy is to choose the card with the largest label in each round. Note that $a_i \geq 2(n + 1 - i)$ for each i since at least one of the cards $2(n + 1 - i), 2(n + 1 - i) + 1, \dots, 2n$ is not chosen in the i th round. As $X + Y = 1 + 2 + \dots + 2n$ is fixed and $X > Y$ (because of this strategy), S is smaller when X is smaller. Thus,

$$S \geq (2 + 4 + \dots + 2n)^2 + (1 + 3 + \dots + (2n - 1))^2.$$

Together with the claim, we deduce

$$S - T \geq \left(\sum_{k=1}^n 2k \right)^2 + \left(\sum_{k=1}^n (2k - 1) \right)^2 - \sum_{k=1}^n (2k - 1)(2k) = N$$

as desired.

Next, we show that B can guarantee that the score of A is at most N . A strategy is to pair up the cards with labels $2m - 1$ and $2m$ for each $m = 1, 2, \dots, n$, and choose a card in the same pair as the card just chosen by A in each round. It is easy to see that

$$T = 1 \cdot 2 + 3 \cdot 4 + \dots + (2n - 1)(2n).$$

Also, as $|a_i - b_i| = 1$ for each i , we have

$$\begin{aligned} S &= X^2 + Y^2 = \frac{(X + Y)^2 + (X - Y)^2}{2} \\ &= \frac{1}{2} \left[(1 + 2 + \dots + 2n)^2 + \left(\sum_{i=1}^n (a_i - b_i) \right)^2 \right] \\ &\leq \frac{(1 + 2 + \dots + 2n)^2 + n^2}{2}. \end{aligned}$$

Equality holds when $a_i - b_i = 1$ for all i or $a_i - b_i = -1$ for all i . This is exactly the case when

$$S - T = \left(\sum_{k=1}^n 2k \right)^2 + \left(\sum_{k=1}^n (2k - 1) \right)^2 - \sum_{k=1}^n (2k - 1)(2k) = N.$$

The proof is complete.

4. Answer: $n = 1$.

When $n = 1$, $(2^1 - 1)(5^1 - 1) = 4$ is a perfect square. Suppose $n > 1$.

- If $n \equiv 3 \pmod{4}$, we have $(2^n - 1)(5^n - 1) \equiv (2)(-1) \equiv 3 \pmod{5}$, which is not true for a square.
- If $n \equiv 1 \pmod{4}$, we rewrite

$$(2^n - 1)(5^n - 1) = (2^n - 1) \cdot 4(5^{n-1} + 5^{n-2} + \cdots + 1).$$

If this is a square, so is $(2^n - 1)(5^{n-1} + 5^{n-2} + \cdots + 1)$. However,

$$(2^n - 1)(5^{n-1} + 5^{n-2} + \cdots + 1) \equiv (-1)(1 + 1 + \cdots + 1) = -n \equiv 3 \pmod{4},$$

which is not true for a square.

- If $n = 2m$ is even, suppose $(2^n - 1)(5^n - 1)$ is a perfect square. Let $2^n - 1 = ds^2$ and $5^n - 1 = dt^2$ where d is squarefree. We first show that $d > 1$. Indeed, if $d = 1$, then $5^{2m} - t^2 = 1$. This gives $(5^m - t)(5^m + t) = 1$, which is impossible as $t > 0$. Now, it is known that the solutions to Pell's equation $x^2 - dy^2 = 1$ are generated by $(x_1 + y_1\sqrt{d})^n = x_n + y_n\sqrt{d}$, where (x_1, y_1) is the fundamental solution.

Claim. Let (x_k, y_k) be a solution to $x^2 - dy^2 = 1$.

- (i) If $k \equiv 0 \pmod{2}$, then $x_k \equiv 1 \pmod{2}$ and $y_k \equiv 0 \pmod{2}$.
- (ii) If $k \equiv 1 \pmod{2}$, then $x_k \equiv x_1 \pmod{2}$ and $y_k \equiv y_1 \pmod{2}$.

Proof.

- (i) Let $k = 2\ell$. Then we have

$$\begin{aligned} x_k + y_k\sqrt{d} &= (x_1 + y_1\sqrt{d})^{2\ell} \\ &= (x_\ell + y_\ell\sqrt{d})^2 \\ &= x_\ell^2 + dy_\ell^2 + 2x_\ell y_\ell\sqrt{d} \\ &= 2x_\ell^2 - 1 + 2x_\ell y_\ell\sqrt{d}. \end{aligned}$$

So $x_k = 2x_\ell^2 - 1 \equiv 1 \pmod{2}$ and $y_k = 2x_\ell y_\ell \equiv 0 \pmod{2}$.

- (ii) Let $k = 2\ell + 1$. By $(x_1 + y_1\sqrt{d})^{2\ell+1} = (x_1 + y_1\sqrt{d})^{2\ell}(x_1 + y_1\sqrt{d})$, we have

$$\begin{aligned} x_{2\ell+1} &= x_{2\ell}x_1 + y_{2\ell}y_1d \equiv x_1 \pmod{2}, \\ y_{2\ell+1} &= x_{2\ell}y_1 + x_1y_{2\ell} \equiv y_1 \pmod{2}. \end{aligned}$$

□

Now, since $2^{2m} - 1 = ds^2$, $(x, y) = (2^m, s)$ is a solution to $x^2 - dy^2 = 1$. By the claim, we must have $x_1 \equiv 0 \pmod{2}$. Similarly, since $5^{2m} - 1 = dt^2$, $(x_k, y_k) = (5^m, t)$ is another solution. By the claim, as $5^m \not\equiv x_1 \pmod{2}$, we must have $k \equiv 0 \pmod{2}$. Let $k = 2\ell$. Then we have $5^m = x_{2\ell} = 2x_\ell^2 - 1$. This implies $2x_\ell^2 - 1 \equiv 0 \pmod{5}$, and hence $x_\ell^2 \equiv 3 \pmod{5}$. This is impossible.

Therefore, the only solution is $n = 1$.